Energy transition study with integration of multiple energy carriers

ESI Final Assignment – Due Mar. 30th, 2026 at 23:59

The goal of this assignment is to practice with modeling the energy conversion, transport and storage technologies learnt in this course, and bring them together in an energy transition study where synergies between the various energy carriers (electricity, heat, hydrogen) are examined. We will build a stylized, 3-zone energy system. Such a system is relatively easy to model, understand and analyze the trade-offs in design alternatives.

We start with the current (also known as brownfield) situation, where energy demand is served by a combination of electricity and natural gas.

To make the designs slightly different, each group of two students is provided with specific data, given in the Excel file group_list.xlsx. The month and year to analyze is unique to each group, and demand multiplier factors to calculate the demand curves are given in the same Excel. The time series for the demands of electricity and heat are given in the file demand_production.xlsx. Initially, heat demand (LT and HT combined) is served by natural gas (NG); note that this will be relevant for the modelling in Part 2.

Part 1: In a remote zone 1 (lat.-long. given to each group), we ask you to choose the installed capacities and model the operation of a mixture of two RES energy conversion technologies (solar and wind), sufficient to supply an electricity demand center in zone 2, a mixed industrial and residential zone. There is no local demand at zone 1.

Dimension the L12 transmission line (in 10 MW increments) to meet the electricity demand, subject to the given grid import constraint in zone 2. Dimension the storage (MW and MWh) needed, and explore the consequences of adding electricity storage at **either** zone 1 or 2. You may assume that storage level is equal to 50% of your chosen total capacity at the start of the study period.

More information:

- Design the system for your given month and year. The CAPEX of RES, transmission line and storage must be annualized to the studied period of 1 month.
- Plan the operation of the energy system for each hourly time-step.
- At each time-step, the demand in zone 1 can import electricity from outside the system under design, until a maximum import limit (MW), at a given price.
- If demand cannot be served from either RES, storage or import, the unserved demand penalty must be accounted for in your annualized cost.

Note that the total cost includes CAPEX, OPEX of technologies as well as the cost of electricity import and unserved demand penalty.

Input data: Month and year for which to analyze the system, location of zone 1, distance between the two zones, demand multiplier factor, techno-economic information about the components, price and limit of electricity import, unserved demand penalty.

All data in Excel/CSV format can be found in the (zipped) 4 files provided in the Assignment folder on Brightspace.

Points to address in the markdown cells of your notebook:

- Explore the trade-off in terms of your chosen KPIs (e.g. total annualized system cost, RES curtailment, utilization factor of the line, unserved demand etc.) between investing in solar/wind production/electricity transmission/storage.
- Consequences of using energy storage -- how does the system design change if storage is located at the production side or at the demand side?
- Address the variables that have most influence on the design, the critical points, limitations and how the design could be improved.
- Support your results with attractive visualizations.

Choose one of your explored design variants to continue to part 2.

Part 2: As part of the zero emission goals of zone 2, the NG infrastructure is dismantled, a H_2 demand is added due to decarbonization of industrial processes, and the original heat demand is split into LT (70%) and HT(30%), where the LT heat demand is to be served by heat pumps and the HT heat demand is to be served by hydrogen boilers. Another remote location, zone 3, is chosen as the site for H_2 production and storage. The H_2 production profile, production price, and maximum allowable tank storage size at zone 3 (due to space constraints) are given in the associated Excel files. Note that penalties apply for unserved heat demand and hydrogen demand, at the same rate of 1500 euro/MWh as for unserved electricity demand.

Dimension the L32 pipeline and the H_2 storage at zone 3 to meet the H_2 and HT heat demands of zone 2. Reinforce renewable production, electricity storage and capacity of the L12 connection to allow for serving the additional LT electricity demand in zone 2 due to the transition to sustainable (residential) heat. You may add additional storage of heat , both for low and high temperatures, at zone 2, as long as you justify your design choices. Next to a zero emission design, same boundary conditions as for Part 1 are applicable.

Once the sizing is done (e.g. via simple heuristics), verify that the operation of the system over 1 month with hourly resolution meets the boundary conditions specified above. This should be done via time-domain simulations. Multiple designs are possible; create at least three diverse designs and evaluate them using the following KPIs:

- Amount of RES energy curtailment
- Amount of unserved electricity, heat or H_2 demand and associated monetary penalties
- Total conversion and transport losses (excluding those for wind and PV production) expressed in absolute units and as percentage of the final energy demand
- Total **additional** annuitized system cost with respect to part 1 (all technologies installed in part 1 are considered fully annuitized for the purpose of part 2)
- Utilization factor of the L12 transmission line
- Number of hours the L32 pipeline is at its maximum capacity and number of hours that the H_2 reservoir is empty or full (binding storage constraints).

Compare the 3 designs and critically analyze them, using the set of KPIs above. Feel free to come up with additional ways to analyze your design, as long as they are justified. Support your results with attractive visualizations.

Reflection on design limitations:

You have designed your system using only 1 month of weather data. Describe *qualitatively* the methodological steps you would take to improve the design of your energy system, given the uncertainties in future weather conditions.

You have designed your system assuming all energy conversion, transport and storage components are perfectly reliable. Describe *qualitatively* the methodological steps you would take to improve the design of your energy system, given the probability that one of these components are temporarily unavailable, due to maintenance or forced outages.

For maximum points, support your reflections with references to the literature.

Choose one of the above designs from part 2 to continue with the bonus points in part 3.

Part 3 (BONUS):

3a) How robust is your chosen design for different monthly weather data and/or different years? Test your design with additional weather data by modifying cell 3 in the code provided. Report on changes in the KPIs due to **different weather** regimes.

3b) Analyze the N-1 situation. With the L12 electricity grid connection of zone 2 outaged, how long can you continue to supply the electricity demand with your chosen electricity storage configuration? Assume that the storage was 50% full at the time of the event and importing from the external grid is constrained. What additional technologies are needed at zone 2 and/or 3, so that the H_2 pipeline can function as a backup for the electricity grid **for 3 days**?

3c) Now do the N-1 analysis the other way round: With the L32 pipeline outaged, how long can you continue to serve the H_2 and HT heat demand with your chosen thermal storage configuration? Assume that the thermal storage was 50% full at the time of the event. What additional technologies are needed at zone 1 and/or 2, so that the L12 transmission line can function as a backup for the H_2 infrastructure **for 3 days**?

How to submit

Upload all your files in *the Assessment->Final Assignment->ESI Assignment* dedicated page on Brightspace before Mar. 30th , 2026, at 23:59. Multiple submissions are possible; we make sure to grade the one with the most recent time stamp. List of files to include:

- the 2 executed Jupyter notebooks for parts 1 and 2, including all results, visualizations and analysis
- optional notebook for part 3 (bonus);
- any additional data files that need be read in order for your code to run;
- AI statement; see template at https://www.uu.nl/en/organisation/ai-policy/students/model-ai-disclosure-statement
- Statement of collaboration, describing how you worked together and which parts of the assignment were done by which members of the group; this can be a separate pdf, or included as a markdown cell at the beginning or end of your notebooks.